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REVIEW ARTICLE

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Unveiling citicoline's mechanisms and clinical relevance in the treatment of neuroinflammatory disorders

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Abstract

Citicoline, a compound produced naturally in small amounts in the human body, assumes a pivotal role in phosphatidylcholine synthesis, a dynamic constituent of membranes of neurons. Across diverse models of brain injury and neurodegeneration, citicoline has demonstrated its potential through neuroprotective and anti-inflammatory effects. This review aims to elucidate citicoline's anti-inflammatory mechanism and its clinical implications in conditions such as ischemic stroke, head trauma, glaucoma, and age-associated memory impairment. Citicoline's anti-inflammatory prowess is rooted in its ability to stabilize cellular membranes,

Abbreviations: Ach, acetylcholine; AD, Alzheimer's disease; AMPK, adenosine monophosphate protein kinase; CAP, cholinergic anti-inflammatory pathway; ChAT, choline acetyltransferase; CNS, central nervous system; CO-OPERA, Citicoline Oral Plus Endovascular Recanalization for Acute Ischemic Stroke; COX-2, cyclooxygenase-2; CRP, C-reactive protein; CTP, cytidine triphosphate; DAG, diacylglycerol; EAE, experimental autoimmune encephalomyelitis; ECM, extracellular matrix; ERK, extracellular signal-regulated kinase; GPx, glutathione peroxidase; GSH, glutathione; ICAM-1, intercellular adhesion molecule-1; ICH, intracerebral hemorrhage; IKK, Iκb kinase; IL, interleukin; iNOS, inducible nitric oxide synthase; IOP, increased intraocular pressure; IRS-1, insulin receptor substrate-1; JAK, janus kinases; JNK, C-Jun N-terminal kinase; LC3-II, light chain protein 3-II; LPS, lipopolysaccharide; MAPK, mitogen-activated protein kinase; MDD, major depressive disorder; MMP-9, matrix metalloproteinase 9; MS, multiple sclerosis; mTOR, mammalian target of rapamycin; NEMO, NF-κB essential modulator; NFTs, neurofibrillary tangles; NF-κB, nuclear factor-kappa B; NIK, NF-κB-inducing kinase; NMDA, N-methyl-D-aspartate; OAG, open-angle glaucoma; OGD/R, oxygen–glucose deprivation and reoxygenation; P38 MAPKs, P38 mitogen-activated protein kinases; PD, Parkinson's disease; PKC, protein kinase C; PPAR-γ, peroxisome proliferator-activated receptor-gamma; RCTs, randomized controlled trials; ROS, reactive oxygen species; SAM, S-adenosylmethionine; SMT, S-methylisothiourea sulfate; SOD, superoxide dismutase; STAT, signal transducer and activator of transcription; TBI, traumatic brain injury; tGCI, transient global cerebral ischemia; tMCAO, transient middle cerebral artery occlusion; TNF-α, tumor necrosis factor-alpha; VCAM-1, vascular cell adhesion molecule-1; VEPs, visual evoked potentials; VFQ-25, visual function questionnaire-25; α7nAChR, Alpha7 nicotinic acetylcholine receptor.

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thereby curbing the excessive release of glutamate—a pro-inflammatory neurotransmitter. Moreover, it actively diminishes free radicals and inflammatory cytokines productions, which could otherwise harm neurons and incite neuroinflammation. It also exhibits the potential to modulate microglia activity, the brain's resident immune cells, and hinder the activation of NF-κB, a transcription factor governing inflammatory genes. Clinical trials have subjected citicoline to rigorous scrutiny in patients grappling with acute ischemic stroke, head trauma, glaucoma, and age-related memory impairment. While findings from these trials are mixed, numerous studies suggest that citicoline could confer improvements in neurological function, disability reduction, expedited recovery, and cognitive decline prevention within these cohorts. Additionally, citicoline boasts a favorable safety profile and high tolerability. In summary, citicoline stands as a promising agent, wielding both neuroprotective and anti-inflammatory potential across a spectrum of neurological conditions. However, further research is imperative to delineate the optimal dosage, treatment duration, and underlying mechanisms. Moreover, identifying specific patient subgroups most likely to reap the benefits of citicoline as a new therapy remains a critical avenue for exploration.

KEYWORDS
cell signaling, citicoline, CNS disorders, neuroinflammation, new therapies

1 | INTRODUCTION

Citicoline is a naturally occurring compound that plays a vital role in the synthesis of phosphatidylcholine, a crucial component of cellular membranes.[1] Phosphatidylcholine plays a vital role in both preserving cellular integrity and function, as well as regulating inflammatory responses. Inflammation is a complex biological process marked by the activation of immune cells and the subsequent release of a range of mediators, such as cytokines, chemokines, and reactive oxygen species (ROS).[2] Inflammation serves a dual role, as it can be advantageous in combating infections and facilitating tissue repair, yet it can also inflict damage when it persists or becomes overly intense. Prolonged inflammation has been linked to the development of numerous diseases, including neurodegenerative disorders, gastrointestinal, pulmonary, cardiovascular, cancer, and diabetes.[3–10]

Citicoline has showcased its capacity to manifest anti-inflammatory properties in various experimental models and clinical settings, as depicted in Figure 1. It possesses the ability to inhibit NF-κB, a crucial transcription factor that regulates the expression of inflammatory genes, resulting in a reduction in the synthesis and release of pro-inflammatory cytokines like tumor necrosis factor-alpha (TNF-α), interleukin-1 beta (IL-1β), and interleukin-6 (IL-6) (Figure 4).[11] Citicoline possesses the ability not only to counteract ROS and enhance the functionality of antioxidant enzymes such as superoxide dismutase (SOD) and glutathione peroxidase (GPx), thereby protecting cells from oxidative stress and inflammation but it can also modulate the activity of microglia, the immune cells residing within the central nervous system (CNS). This modulation prevents their overactivation and the consequent neurotoxic effects.[12] Additionally, citicoline can mitigate the entry of peripheral immune cells into the CNS and decrease the expression of adhesion molecules like intercellular adhesion molecule-1 (ICAM-1) and vascular cell adhesion molecule-1 (VCAM-1), which facilitate the migration of leukocytes across the blood–brain barrier (BBB).[13]

Citicoline's anti-inflammatory attributes have been validated through a variety of preclinical and clinical studies covering a wide spectrum of inflammatory conditions.[14] As an illustration, citicoline has demonstrated its effectiveness in diminishing brain edema, mitigating neuronal damage, and alleviating neurological impairments in animal models of ischemic stroke, traumatic brain injury (TBI), and spinal cord injury. Moreover, citicoline has exhibited positive outcomes by enhancing cognitive function and mood in individuals with mild cognitive impairment, Alzheimer's disease (AD), and depression, all of which are conditions linked to chronic neuroinflammation.[15] Additionally, citicoline has been observed to lower circulating levels of C-reactive protein (CRP), which serves as an indicator of systemic inflammation. It has also been found to improve endothelial function in individuals with vascular risk factors or cardiovascular diseases.[16]

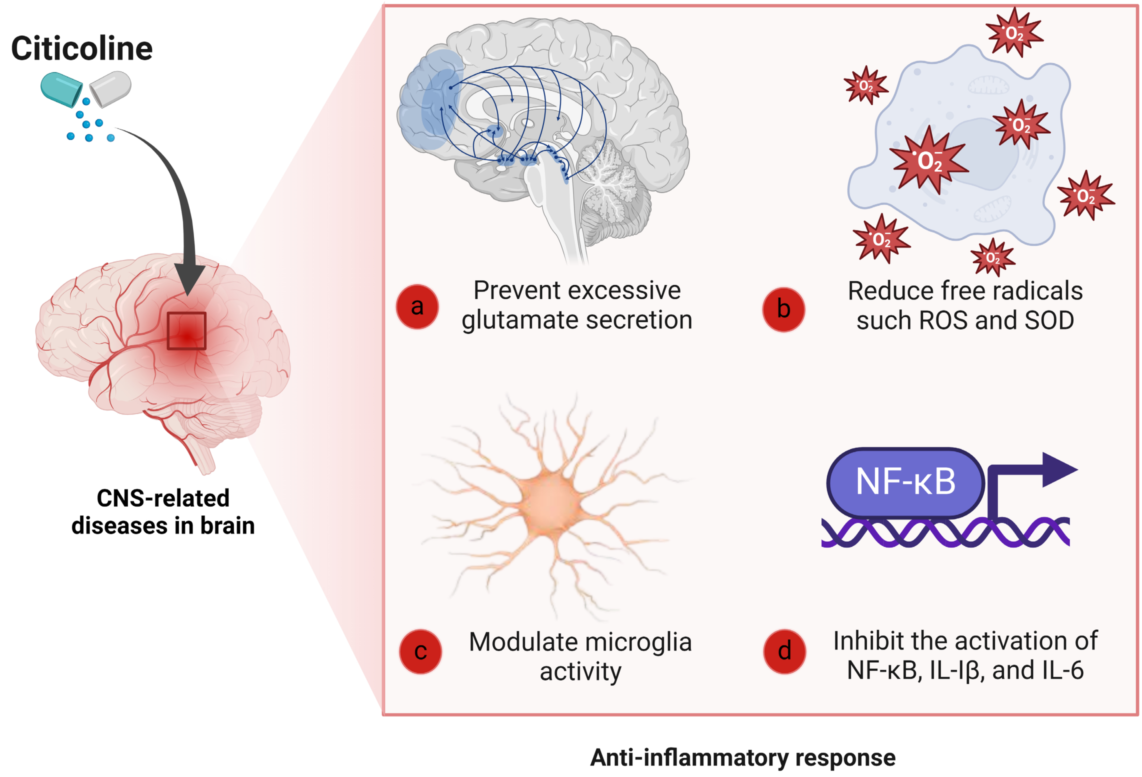

FIGURE 1 Protective effects of citicoline. Citicoline demonstrates its anti-inflammatory prowess through a multifaceted approach. It inhibits the activation of NF-κB, leading to reduced production of pro-inflammatory cytokines like TNF-α, IL-1β, and IL-6. Furthermore, it serves as a scavenger for ROS, enhances the activity of antioxidant enzymes like SOD and GPx, moderates microglial function, and restricts the entry of peripheral immune cells into the CNS by reducing the expression of adhesion molecules such as ICAM-1 and VCAM-1.

2 | PATHOPHYSIOLOGICAL MECHANISMS OF CHRONIC NEUROINFLAMMATION

Chronic neuroinflammation is a complex process that involves the persistent activation of the CNS immune cells, primarily microglia, and astrocytes, in response to various pathological stimuli. This sustained inflammatory response is implicated in a wide range of neurological disorders, including neurodegenerative diseases, psychiatric conditions, and traumatic brain injuries.

2.1 | Microglial activation and function

Microglia are the resident immune cells of the CNS and play a pivotal role in maintaining brain homeostasis. Under normal conditions, microglia exhibit a ramified morphology and continuously monitor the neural environment. However, upon exposure to harmful stimuli such as pathogens, injury, or stress, microglia become activated.[17] This activation is characterized by a transformation into an amoeboid shape, increased proliferation, and the release of pro-inflammatory cytokines, chemokines, and ROS.[18] While acute microglial activation can be protective, chronic activation leads to detrimental effects.[19] Activated microglia release inflammatory mediators such as TNF-α, IL-1β, and IL-6, which can induce neuronal damage, disrupt synaptic plasticity, and impair cognitive functions.[20] This chronic inflammatory state can result in a feedback loop, where the released cytokines further exacerbate microglial activation, perpetuating the cycle of inflammation and tissue damage.

2.2 | Astrocytic response

Astrocytes, another type of glial cell, also play a crucial role in neuroinflammation. In response to inflammatory signals, astrocytes become reactive, a process characterized by changes in morphology and the upregulation of glial fibrillary acidic protein.[21] Reactive astrocytes contribute to neuroinflammation by releasing pro-inflammatory cytokines and chemokines, which can further activate microglia and promote neuroinflammatory processes.[22] Moreover, astrocytes are involved in maintaining the integrity of the BBB. Chronic neuroinflammation can lead to BBB disruption, allowing peripheral immune cells and inflammatory mediators to infiltrate the CNS, exacerbating the inflammatory response and contributing to neurodegeneration.[23]

2.3 | Role of cytokines and chemokines

During neuroinflammation, cytokines and chemokines, such as CCL2 (MCP-1) and CXCL10 (IP-10) are released by activated microglia and astrocytes in the CNS.[24] These cells respond to injury or infection by producing signaling molecules that recruit additional immune cells, including macrophages and lymphocytes, to the affected area. The interaction between cytokines and their receptors on immune cells leads to a cascade of inflammatory responses. Cytokines like TNF-α can disrupt the integrity of the BBB, allowing immune cells to infiltrate the CNS. Additionally, pro-inflammatory cytokines can lead to neuronal apoptosis and contribute to the pathogenesis of neurodegenerative diseases.[25]

While cytokines and chemokines initiate and propagate inflammation, there are also anti-inflammatory cytokines, such as IL-10 and transforming growth factor-beta (TGF-β), that help to resolve inflammation and promote tissue repair.[26] The balance between pro-inflammatory and anti-inflammatory signals is crucial for maintaining homeostasis in the CNS. Dysregulation can lead to chronic inflammation, contributing to neuroinflammatory conditions.[27]

2.4 | Oxidative stress and neuroinflammation

Chronic neuroinflammation is closely linked to oxidative stress, a condition characterized by an imbalance between ROS production and antioxidant defenses. Activated microglia and astrocytes produce ROS, which can damage cellular components, leading to neuronal injury and death. This oxidative stress can further stimulate neuroinflammatory pathways, creating a vicious cycle that perpetuates neurodegeneration.[28]

3 | THE ROLE OF NEUROINFLAMMATION IN DIFFERENT NEUROLOGICAL DISORDERS

Neuroinflammation is increasingly recognized as a critical factor in the pathogenesis of various neurological disorders. It refers to the inflammatory response within the CNS, which includes the brain and spinal cord. This response can be triggered by a variety of factors, including infections, injuries, and neurodegenerative processes. Understanding the role of neuroinflammation is essential for developing effective therapeutic strategies for conditions such as Alzheimer's disease, Parkinson's disease, multiple sclerosis (MS), and others.

4 | NEUROINFLAMMATION IN NEURODEGENERATIVE DISEASES

Neurodegenerative diseases are characterized by progressive loss of structure and function of neurons, and neuroinflammation plays a critical role in these conditions. In AD, neuroinflammation is initiated by the accumulation of amyloid-beta plaques and tau tangles. These protein aggregates activate microglia and astrocytes, leading to the release of pro-inflammatory cytokines such as IL-6 and TNF-α.[29] This inflammatory environment contributes to neuronal damage and cognitive decline, suggesting that targeting neuroinflammatory pathways could be a therapeutic strategy. Similar to AD, PD is associated with the accumulation of misfolded proteins, particularly alpha-synuclein.[30] Neuroinflammation in PD is characterized by the activation of microglia, which can exacerbate neuronal loss through the release of inflammatory mediators.[31] Studies suggest that chronic neuroinflammation may precede the clinical onset of PD, indicating its potential role in disease initiation.[32]

MS is a demyelinating disease where neuroinflammation is a hallmark feature. In MS, autoreactive T cells and B cells infiltrate the CNS, leading to the destruction of myelin sheaths surrounding neurons.[33] This process is accompanied by the activation of microglia and the release of inflammatory cytokines, which further perpetuate the inflammatory cycle.[34] Neuroinflammation in MS is associated with both acute relapses and chronic progression of the disease, making it a critical target for therapeutic intervention.[35]

In conditions such as TBI and stroke, neuroinflammation plays a significant role in the secondary injury processes that follow the initial insult. Following TBI, a cascade of inflammatory responses is triggered, involving the activation of microglia and the release of cytokines and chemokines. This neuroinflammatory response can lead to further neuronal damage and contribute to long-term cognitive and functional impairments.[36] Stroke induces a rapid inflammatory response that can exacerbate brain injury. The activation of immune cells and the release of inflammatory mediators can lead to neuronal death and worsen outcomes. Modulating the inflammatory response post-stroke is an area of active research, with the goal of improving recovery and reducing disability.[37]

Inflammation can alter neuronal excitability and contribute to the development of epilepsy. Neuroinflammatory processes may be involved in both the initiation and progression of seizure disorders.[38] Additionally, there is growing evidence linking neuroinflammation to mood disorders. Elevated levels of inflammatory cytokines have been observed in patients with depression, suggesting that inflammation may play a role in the pathophysiology of these conditions.[39] Furthermore, conditions such as autism spectrum disorder and attention-deficit/hyperactivity disorder (ADHD) have also been associated with neuroinflammatory processes, although the exact mechanisms remain under investigation.[40]

5 | MECHANISMS OF ACTION OF CITICOLINE

Citicoline exerts a vital role in the maintenance and synthesis of phospholipids, especially phosphatidylcholine, in the brain.[1] Phosphatidylcholine is a main component of cell membranes and is believed to be essential for neuronal function and integrity. Citicoline is also involved in the production of acetylcholine (Ach), a neurotransmitter that mediates cognitive processes such as memory, attention, and learning.

Citicoline is derived from choline and cytidine, two precursors that are obtained from dietary sources or endogenous metabolism.[41] It crosses the BBB and enters the brain tissue, where it is metabolized into choline and cytidine by hydrolysis. Choline and cytidine can then be recombined to form citicoline again or be used for other purposes. Choline can be converted into Ach by choline acetyltransferase (ChAT), or into phosphatidylcholine by phosphocholine cytidylyltransferase. Cytidine can be converted into uridine by cytidine deaminase, or into cytidine triphosphate (CTP) by cytidine kinase. CTP is a key intermediate for phosphatidylcholine and other phospholipids synthesis.[42]

6 | ANTI-INFLAMMATORY AND NEUROPROTECTIVE MECHANISMS OF CITICOLINE

6.1 | Effect on glutamate neurotransmitter

One suggested mechanism underlying the action of citicoline involves its modulation of N-methyl-D-aspartate (NMDA) receptors and the regulation of glutamate neurotransmission. NMDA receptors are ionotropic glutamate receptors responsible for mediating excitatory synaptic transmission and playing crucial roles in processes such as learning, memory, and synaptic plasticity. However, excessive activation of NMDA receptors can lead to excitotoxicity, a process that contributes to neuronal death in many neurological disorders.[43] Citicoline may protect neurons from excitotoxicity by reducing the influx of calcium through NMDA receptors, enhancing the activity of glutamate transporters, and increasing the synthesis of glutathione (GSH), an antioxidant molecule.[44] Furthermore, citicoline has the potential to boost the restoration of neuronal function by promoting the expression of neurotrophic factors, notably brain-derived neurotrophic factor (BDNF), and facilitating the development of new synaptic connections.[45] These effects could offer an explanation for the favorable impact of citicoline on cognitive function and neurological outcomes observed in individuals with conditions such as stroke, TBI, AD, and various other neurodegenerative disorders.

Numerous preclinical studies have reported the neuroprotective effects of citicoline in various models of brain injury. Citicoline has been shown to decrease infarct volume and improve neurological outcomes in animal models of ischemic stroke. It does this by enhancing phosphatidylcholine synthesis, which is essential for cell membrane integrity and repair, in addition to modulation of glutamate transport.[46] In models of TBI, citicoline has demonstrated protective effects against both white and gray matter damage, suggesting its potential utility in acute brain injuries.[47–49] Additionally, citicoline was found to protect motor neurons against apoptotic changes in a model of chronic glutamate excitotoxicity in vitro.[50] In vitro studies using cultured rat cortical neurons exposed to oxygen–glucose deprivation, demonstrated that citicoline could prevent cell death and reduce glutamate release, supporting its role in protecting neuronal integrity during metabolic stress.[51]

6.2 | Effect on synthesis of phosphatidylcholine

Citicoline is derived from cytidine and choline, two molecules that are involved in the biosynthesis of phosphatidylcholine.[52] Citicoline can be produced endogenously in the body or obtained exogenously from dietary sources or supplements. Once citicoline enters the brain, it undergoes hydrolysis, breaking down into cytidine and choline. These components can then traverse the BBB and enter neurons. There, they are recombined into citicoline, which acts as a precursor for phosphatidylcholine synthesis.[53]

There are several mechanisms by which citicoline increases phosphatidylcholine formation in the brain (Figure 2). One of them is by converting 1, 2-diacylglycerol

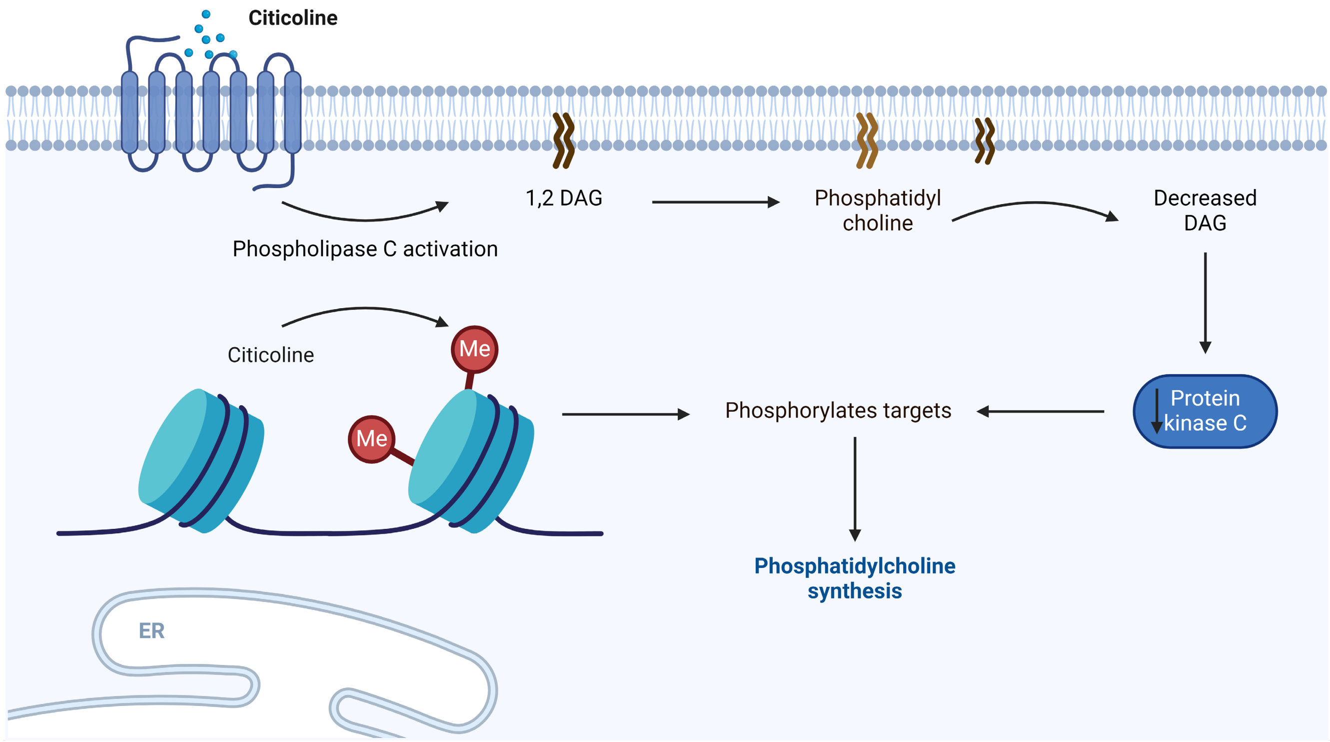

FIGURE 2 Effect on phosphatidylcholine synthesis. The mechanism involves the conversion of DAG, a byproduct of phospholipase C activation, into phosphatidylcholine. This action prevents DAG accumulation and reduces the activation of PKC, a signaling molecule known to trigger cell death and inflammation. Another mechanism involves Citicoline donating methyl groups to SAM, a cofactor necessary for methylation reactions essential for phosphatidylcholine synthesis.

(DAG), a byproduct of phospholipase C activation, into phosphatidylcholine. This prevents DAG accumulation and reduces the activation of protein kinase C (PKC), a signaling molecule that can trigger cell death and inflammation.[54] Another mechanism is by donating methyl groups to S-adenosylmethionine (SAM), a cofactor for methylation reactions that are required for phosphatidylcholine synthesis.[55]

By increasing phosphatidylcholine synthesis, citicoline can enhance the production and release of Ach, a neurotransmitter that is involved in memory, learning, attention, and other cognitive functions.[56] Citicoline can also protect neurons from oxidative stress, excitotoxicity, inflammation, and apoptosis by stabilizing cell membranes and modulating membrane fluidity. Furthermore, citicoline can improve cerebral blood flow and glucose metabolism by increasing nitric oxide (NO) production and reducing vascular resistance.[57]

Preclinical studies have shown that citicoline treatment leads to an increase in the incorporation of labeled precursors into phospholipids, particularly phosphatidylcholine, in brain tissues. This effect is particularly pronounced in models of hypoglycemia-induced neuronal death, where phosphatidylcholine levels typically decline due to membrane degradation.[58] Additionally, in an experimental model of cerebral ischemia, citicoline administration was associated with a significant increase in phosphatidylcholine content and a decrease in free fatty acids, indicating a restoration of lipid metabolism disrupted by ischemia.[59] Another study demonstrated that citicoline could counteract the effects of hypoxia by enhancing the synthesis of structural phospholipids, thereby mitigating the loss of neuronal integrity and function.[60]

6.3 | Effect on pro-inflammatory cytokines

Citicoline has been documented to regulate pro-inflammatory cytokines. Raised levels of pro-inflammatory cytokines, including TNF-α, IL-1β, and IL-6, are commonly observed in numerous neurological disorders and play a role in causing neuronal damage, oxidative stress, and cognitive impairment.[61]

Multiple studies have put forward the idea that citicoline could exert anti-inflammatory effects by diminishing the generation and release of pro-inflammatory cytokines in various models of brain injury and neurodegenerative conditions.[62] For example, citicoline has been documented to lower TNF-α, IL-1β, and IL-6 levels in both the brain and plasma of rats that have undergone ischemic stroke or TBI.[61] Citicoline also declined pro-inflammatory genes and protein expression in cultured microglia and astrocytes stimulated with lipopolysaccharide (LPS), a bacterial endotoxin that induces inflammation.

Citicoline's interaction with microglial receptors has been highlighted in studies examining its neuroprotective mechanisms. Specifically, it has been shown to interact with *P2Y6 receptors* on microglia. The activation of these receptors is believed to play a crucial role in mediating the neuroprotective effects of citicoline, particularly in the context of ischemic brain injury. The administration of citicoline before ischemic events has been shown to enhance its protective effects, which can be blocked by P2Y6 receptor antagonists, indicating that citicoline's action involves receptor-mediated pathways in vivo.[63]

In other inflammatory models, citicoline showed neuroprotective effects via reducing inflammatory cytokines levels. In AD, citicoline administration significantly decreased levels of pro-inflammatory cytokines like MIP-1α, TNFα, IL-1β, and MCP-1 in an in vivo study. The cytidine component of citicoline specifically reduced IL-6, RANTES, and anti-inflammatory cytokine IL-10. Additionally, Citicoline triggers upregulation of SIRT1 expression, which reduces neuronal inflammatory response in hyperinflammatory conditions.[64,65] Moreover, research indicates that citicoline may help in managing PD by enhancing dopamine levels and reducing neuroinflammation. It has been shown to improve cognitive functions and reduce symptoms associated with PD, potentially through its anti-inflammatory properties.[66] Citicoline also has been evaluated in rodent models of TBI, where it demonstrated neuroprotective effects by activating microglia, reducing neuronal apoptosis, and promoting neuronal proliferation. These actions are associated with decreased levels of pro-inflammatory cytokines, suggesting a reduction in neuroinflammation.[67] Furthermore, citicoline is often used in stroke models, where it has been reported to reduce infarct volume and improve neurological outcomes. Its mechanism includes the inhibition of phospholipase A2, which is involved in neuroinflammation (Figure 3), thereby decreasing the levels of ROS and pro-inflammatory cytokines.[68]

Citicoline's ability to inhibit pro-inflammatory cytokines may be attributed to various mechanisms. These mechanisms include the inhibition of NF-κB activation, a transcription factor responsible for regulating pro-inflammatory gene expression; the enhancement of peroxisome proliferator-activated receptor-gamma (PPAR-γ) activity, a nuclear receptor known for its anti-inflammatory properties; and the elevation of Ach levels, a neurotransmitter that plays a role in modulating immune responses.[69] The modulation of pro-inflammatory cytokines by citicoline may have beneficial effects on neuronal survival, synaptic plasticity, and cognitive function. For instance, citicoline has been shown to diminish neuronal apoptosis, improve neurogenesis, and repair synaptic transmission and memory in brain injury and neurodegeneration animal models (Table 1).[70]

6.4 | Effect on NF-κB pathway

One conceivable mechanism through which citicoline exerts its anti-inflammatory effect involves the modulation of the NF-κB pathway, which plays a central role in regulating inflammation, cell survival, and gene expression.[71–74] NF-κB consists of a family of transcription factors that can be activated by various stimuli, such as cytokines, oxidative stress, and pathogens. Upon activation, NF-κB dimers translocate from the cytoplasm to the nucleus, where they bind to specific DNA sequences and initiate the expression of target genes related to inflammatory and immune responses.[75]

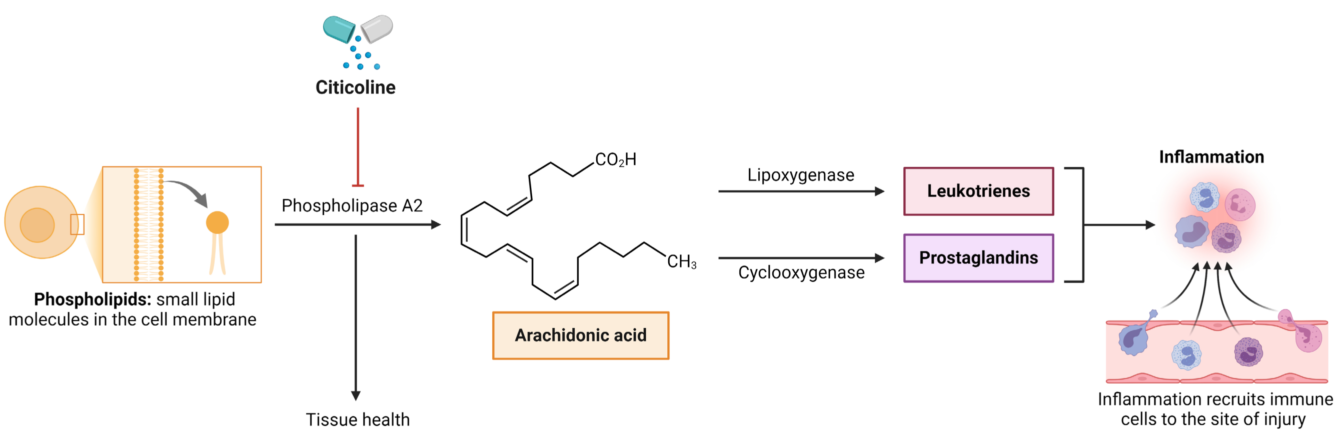

FIGURE 3 Effect of citicoline on COX pathway. A schematic representation illustrating how citicoline inhibits phospholipase activity, potentially leading to reduced production of COX-2, lipoxygenase, and inflammatory mediators, improved cellular membrane integrity, and enhanced neuroprotective effect.

TABLE 1 Molecular neuroprotective mechanism of citicoline.

Target pathway	Model	Mechanism of citicoline
Glutamate neurotransmitter	• Stroke • TBI • AD	• Modulating NMDA receptors • Reducing calcium influx • Enhancing glutamate transporter activity • Increasing glutathione synthesis • Promoting neurotrophic factor expression • Facilitating synaptic connections
Phosphatidylcholine synthesis	• Brain injury • Stroke • Hypoglycemia • Cerebral ischemia • Hypoxia	• Converting 1, 2-DAG to phosphatidylcholine • Donating methyl groups to SAM for phosphatidylcholine synthesis • Enhancing Ach production • Protecting neurons from oxidative stress, excitotoxicity, inflammation, apoptosis • Improving cerebral blood flow and glucose metabolism
Pro-inflammatory cytokines	• Brain injury • AD • PD • TBI • Stroke	• Reducing levels of TNF-α, IL-1β, IL-6, interacting with P2Y6 receptors on microglia • Enhancing SIRT1 expression • Reducing neuronal inflammatory response • Reducing neuronal apoptosis • Promoting neuronal proliferation • Inhibiting phospholipase A2 • Decreasing ROS
NF-κB	• Cerebral ischemia–reperfusion injury • Glutamate excitotoxicity	• Inhibiting NF-κB activation • Reducing pro-inflammatory cytokine production • Interacting with NEMO, IKKs, IκBs, or NF-κB dimers
COX-2	• Cancer	• Binding to the 20S proteasome modulates its activity • Reducing COX-2 expression by decreasing phospholipase A2 stimulation
iNOS	• Diabetic neuropathy • Glaucoma	• Inhibiting iNOS activity • Reducing oxidative stress • Enhancing nerve regeneration
MMP-9	• Ischemic stroke • Ocular damage	• Lowering MMP-9 levels and activity • Increasing TGF-β1 expression • Promoting neuroprotection and axonal regeneration by modulating MMP activity • Modulating Ach receptors to regulate MMP-9 expression and function
MAPK pathway	• Focal cerebral ischemia • Human brain microvessel endothelial cells	• Inhibiting p38 MAPK and JNK activation • Stimulating ERK phosphorylation • Increasing IRS-1 expression • Enhancing angiogenesis, wound healing, and survival of endothelial cells • Decreasing microglial activation and astrocytic proliferation
STAT pathway	• AD	• Inhibiting JAK2 and STAT3 activation • Decreasing pro-inflammatory cytokine production
CAP	• AD	• Increasing Ach levels by increasing ChAT and α7nAChR expression • Stimulating the release of Ach, and directly activates α7nAChR to reduce inflammation and oxidative stress • Improving neurogenesis, synaptic plasticity, and neurotransmission
Apoptosis	• Ocular hypertension • Diabetic neuropathy • Transient focal ischemia	• Reducing caspase-3 and caspase-9 levels • Lowering pro-apoptotic protein expression • Increasing anti-apoptotic protein expression • Reducing caspase-8 activation • Promoting the expression of neurotrophic factors and neurite regeneration
Autophagy	• AD	• Activating AMPK and Sirtuin1, • Inhibiting mTOR and increasing beclin-1 levels • Increasing LC3-II • Decreasing p62 levels • Increasing Beclin-1 and Atg5 expression

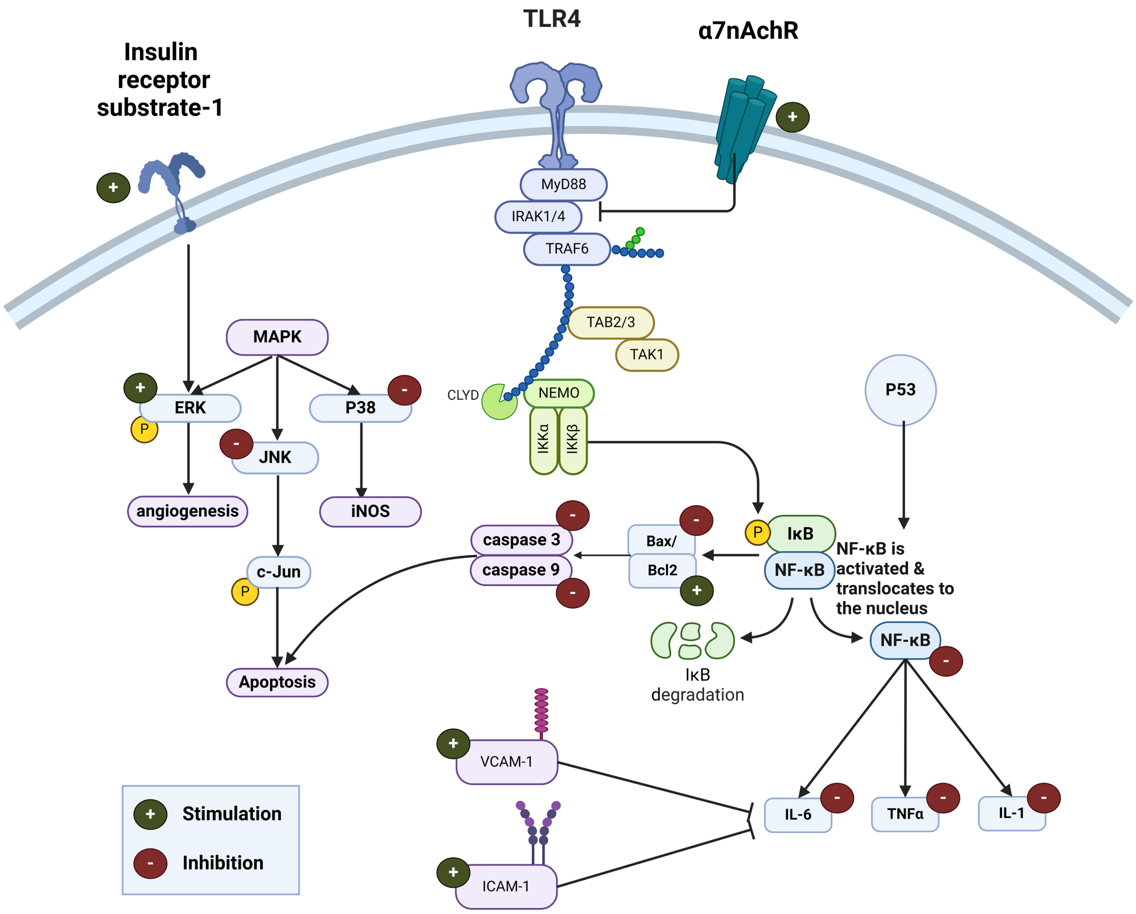

FIGURE 4 Molecular mechanisms of anti-inflammatory effects of citicoline. Citicoline shows a wide spectrum of neuroprotective and neuroregenerative attributes, and exerts a profound influence across a network of interconnected molecular pathways within the body. It takes center stage in modulating vital ERK, JNK, and MAPK signaling pathways that underpin cell growth, differentiation, and survival, thus enhancing cellular resilience and fostering holistic neuroprotection. Furthermore, citicoline actively promotes angiogenesis, facilitating heightened blood circulation and nutrient delivery, especially within the brain, thereby fortifying tissue repair and the path to recovery. Its pivotal role in the regulation of apoptosis becomes evident as it fine-tunes the equilibrium between Bax) and Bcl-2 factors, safeguarding cells from premature demise and reinforcing their survival. Citicoline also exerts a significant impact on the caspase-3 and caspase-9 activity, the key player enzymes orchestrating programmed cell death, thereby mitigating cellular attrition and augmenting cellular longevity. Beyond this, Citicoline's anti-inflammatory properties extend their reach to encompass NF-κB modulation, effectively countering the adverse effects of inflammation. Simultaneously, it restricts the production of the inflammatory cytokines, actively participating in the dampening of inflammatory responses. The intricate interplay between Citicoline and the α7 nAChR influences a diverse array of neuroprotective mechanisms, including synaptic plasticity and the modulation of neuroinflammation, thus contributing to its cognitive enhancement and overall neuroprotective effects.

The NF-κB pathway can be categorized into two primary branches: the canonical and non-canonical pathways. The canonical pathway is typically activated by a broad range of stimuli and involves the degradation of inhibitory proteins referred to as IκBs by the IκB kinase (IKK) complex. This complex comprises two catalytic subunits, IKKα and IKKβ, in addition to a regulatory subunit named NF-κB essential modulator (NEMO). Conversely, the non-canonical pathway is triggered by specific stimuli, such as particular members of the TNF-α receptor superfamily. This pathway involves the conversion of a precursor protein, p100, into p52, a process facilitated by the NF-κB-inducing kinase (NIK) and IKKα.

Both of these pathways have the capacity to interact with each other through positive or negative feedback loops. This interaction results in a complex interplay that determines the specificity and duration of NF-κB activation. When NF-κB signaling becomes dysregulated, it can consequently give rise to chronic inflammation, tissue damage, and contribute to the development of various diseases, including cancer, autoimmune disorders, and neurodegenerative conditions.[76]

Several studies have suggested that citicoline possesses the ability to regulate NF-κB signaling in a variety of cell types and under different circumstances. To illustrate, citicoline has been shown to inhibit NF-κB activation and reduce the production of pro-inflammatory cytokines in microglial cells exposed to LPS, a bacterial endotoxin that mimics infection. Moreover, citicoline has been proven to suppress NF-κB activation and mitigate neuronal apoptosis in rat models of cerebral ischemia–reperfusion injury. Additionally, citicoline has been noted to enhance NF-κB activation, thereby promoting neuronal survival in models characterized by glutamate excitotoxicity and oxidative stress.[77]

These discoveries suggest that citicoline's impact on NF-κB signaling varies depending on the circumstances and the nature of the stimulus. In response to detrimental triggers like LPS or ischemia–reperfusion injury, citicoline might function as an anti-inflammatory agent by suppressing NF-κB activation.[12] Conversely, citicoline could potentially serve as a neuroprotective agent when it comes to promoting NF-κB activation in response to favorable stimuli, such as glutamate or oxidative stress. The precise mechanisms through which citicoline influences NF-κB signaling remain somewhat enigmatic; however, they may encompass direct or indirect interactions with elements of the NF-κB pathway, including NEMO, IKKs, IκBs, or NF-κB dimers.[78]

6.5 | Effect on cyclooxygenase-2

The impact of citicoline on cyclooxygenase-2 (COX-2), an enzyme that is involved in inflammation and cancer, is not well understood. COX-2 is a key target for anti-inflammatory and anti-cancer drugs, but it also has some beneficial roles in the body. COX-inhibitors can reduce pain, fever, and swelling, but they can also increase the risk of heart attacks, stroke, and gastrointestinal bleeding. Therefore, finding drugs that can modulate COX-2 activity in a selective and safe way is a major challenge.[79]

6.6 | Citicoline binds and modulates 20S proteasome activity

The 20S proteasome is a cluster of proteases responsible for breaking down proteins that are no longer needed or have become damaged within the cell. This process is vital for sustaining protein balance and overseeing a range of cellular functions. Importantly, the 20S proteasome is capable of degrading COX-2, thereby regulating its levels and activity.[80] A recent study conducted by Sbardella et al. in 2020 revealed that citicoline has the capability to bind to multiple sites on the 20S proteasome, thereby influencing its proteolytic function. Citicoline exhibits a bimodal allosteric modulation effect, meaning it can either enhance or inhibit the activity of the 20S proteasome depending on the specific substrate and concentration used. The authors of the study proposed that citicoline might induce a structural alteration in the 20S proteasome, impacting its catalytic efficiency. Additionally, the research demonstrated that citicoline has the ability to modify the composition of assembled proteasome populations and the turnover of ubiquitinated proteins in human neuroblastoma cells.[81] Ubiquitination is a process that scripts proteins for degradation by the 20S proteasome. Citicoline may affect the balance between protein synthesis and degradation, thus influencing protein quality and quantity.[82] Moreover, Citicoline has been shown to reduce COX-2 expression by decreasing the stimulation of phospholipase A2. By inhibiting phospholipase A2, citicoline can reduce the production of COX-2, lipoxygenase, and hydroxyl radicals, which are harmful to cells, and prevent the degradation of cardiolipin, a phospholipid that is important for mitochondrial function (Figure 3).[68,83]

6.7 | Citicoline may have anti-inflammatory and anti-cancer effects via COX-2 regulation

The modulation of 20S proteasome activity by citicoline may have important consequences for COX-2 regulation and function. By affecting the degradation of COX-2, citicoline may alter its expression and activity in different tissues and conditions. COX-2 is often overexpressed in many types of cancers, where it promotes tumor growth, survival, angiogenesis, invasion, metastasis, and resistance to therapy.[82] COX-2 also mediates inflammation, which is a major factor in cancer development and progression. Therefore, inhibiting COX-2 may have anti-inflammatory and anti-cancer effects. However, COX-2 also has some protective roles in the body, such as maintaining vascular integrity, wound healing, renal function, and gastric mucosal defense. Therefore, inhibiting COX-2 may have adverse effects on cardiovascular, renal, and gastrointestinal health.[84] Citicoline may offer a novel strategy to modulate COX-2 activity in a selective and safe way. By binding to the 20S proteasome, citicoline may enhance or inhibit COX-2 degradation depending on the context and the dose. This may allow citicoline to exert anti-inflammatory and anti-cancer effects without compromising the beneficial functions of COX-2.[82] however, there is limited evidence of neuroprotective effects of citicoline via targeting COX-2 enzyme.

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6.8 | Effect on iNOS

Citicoline has been shown to exert an influence on inducible nitric oxide synthase (iNOS), which is an enzyme responsible for generating NO. NO serves as a signaling molecule with various functions in processes such as inflammation, apoptosis, and vascular regulation.[85] In a study conducted by Ahlawat and colleagues, the researchers explored the effects of citicoline and S-methylisothiourea sulfate (SMT), a selective inhibitor of iNOS, both separately and in combination, on neuropathic pain induced by type II diabetes mellitus in rats. Neuropathic pain is a common complication of diabetes, resulting from nerve damage triggered by factors like hyperglycemia, oxidative stress, and inflammation. In their study, Ahlawat and colleagues made the noteworthy discovery that both citicoline and SMT, whether administered individually or in combination, significantly mitigated diabetic neuropathic pain. They gauged this improvement using various metrics, including nerve conduction velocity, mechanical and thermal hyperalgesia, and cold allodynia. Additionally, both citicoline and SMT were observed to reduce oxidative stress and axonal degeneration in the sciatic nerve of diabetic rats. The researchers concluded that the concurrent use of citicoline and SMT may hold promise as a potential therapeutic approach for diabetic neuropathic pain, as it involves the inhibition of iNOS and the enhancement of nerve regeneration.[86]

In a separate investigation conducted by Rossetti and colleagues, they explored the effects of an oral citicoline solution on the quality of life among individuals suffering from chronic open-angle glaucoma (OAG). Chronic OAG is a progressive optic nerve disorder characterized by elevated intraocular pressure (IOP), the death of retinal ganglion cells, and deterioration in visual field function. The study followed a randomized, double-masked, placebo-controlled, cross-over design, where patients were randomly assigned to receive either a 500 mg/day oral citicoline solution or a placebo for a duration of 3 months. Subsequently, the participants switched to the alternative treatment for another 3 months. The primary objective of this study was to evaluate the average change in the intra-patient composite score of the Visual Function Questionnaire-25 (VFQ-25), a tool used to assess the quality of life related to vision. The findings indicated that the oral citicoline solution resulted in a noteworthy enhancement of the VFQ-25 composite score when compared to the placebo, as assessed at the 6-month point relative to the baseline. Particularly, the positive impact of citicoline was more pronounced in individuals whose vision-related quality of life had been more severely affected by glaucoma initially. The authors suggested that citicoline may improve the quality of life in people with glaucoma by modulating the expression and activity of iNOS in the retina and optic nerve.[87]

6.9 | Effect on matrix metalloproteinase 9

Matrix metalloproteinase 9 (MMP-9) is a member of the enzyme family tasked with the degradation of the extracellular matrix (ECM), a complex network of proteins and polysaccharides that offer structural support to cells. MMP-9 plays a pivotal role in numerous physiological processes, including wound healing, angiogenesis (the creation of new blood vessels), and tissue remodeling. However, when its activity becomes dysregulated, MMP-9 can also contribute to pathological conditions such as inflammation, infection, cancer, and neurodegeneration.[88] MMP-9 can degrade various ECM components, such as collagen, elastin, and laminin, as well as release growth factors, cytokines, and chemokines that can affect cell behavior and signaling.[89]

Citicoline has demonstrated the ability to regulate both the expression and activity of MMP-9 in various models of tissue injury. For instance, citicoline has been found to lower MMP-9 levels and activity in the brains of rats experiencing ischemic stroke, in nerve damage, and in the retinas of rats with ocular damage.

In a study on rats with acute ischemic stroke induced by middle cerebral artery occlusion (MCAO), citicoline administration reduced plasma and brain levels of MMP-9.[90–92] Additionally, citicoline increased the expression of MMP-2 and TGF-β1and improved the thickness of the sclera tissue of the rat model of myopia.[93] Furthermore, citicoline provided neuroprotection in peripheral nerve injury via modulating MMP activity and promoting the expression of TIMP-1 to stimulate axonal regeneration.[90] Citicoline also reduced the levels of MMP-9 in stroke and declined their role in enhancing BBB damage and neuronal injury in the early times after injury.[94]

The precise mechanisms through which citicoline regulates MMP-9 are not completely elucidated, but there are proposed pathways. One possibility is that citicoline may inhibit NF-kB, a transcription factor that governs MMP-9 gene expression. Another potential mechanism is the modulation of Ach receptors, which can impact MMP-9 activity. These pathways offer potential insights into how citicoline may exert its regulatory effects on MMP-9 expression and function.[95] The effect of citicoline on MMP-9 may have therapeutic implications for various diseases that involve ECM degradation and inflammation. Citicoline has been used clinically for the treatment of cognitive impairment, stroke, TBI, and glaucoma, among other conditions.[96–98]

6.10 | Effect on MAPK pathway

Phosphatidylcholine plays a pivotal role in the regulation of multiple intracellular signaling pathways, as illustrated in Figure 4. One of these pathways is the mitogen-activated protein kinase (MAPK) family, which encompasses a group of serine/threonine kinases responsible for mediating cellular responses to a range of stimuli, including stress, inflammation, and apoptosis. Remarkably, research has shown that citicoline effectively inhibits the activation of MAPKs, particularly p38 MAPK and c-Jun N-terminal kinase (JNK), in the aftermath of focal cerebral ischemia in rats. This observation suggests that citicoline may exert a regulatory influence on these critical signaling pathways in response to brain injury.[65] This inhibition of MAPK activation by citicoline could potentially lead to a decrease in the expression of pro-inflammatory cytokines and other agents responsible for neuronal damage. It is important to note that these substances can have either neuroprotective or neurodegenerative effects, depending on the specific context in which they are operating.[99] Furthermore, citicoline might also influence the activity of other MAPKs, including extracellular signal-regulated kinase (ERK). As a result, citicoline could hold promise for potential therapeutic applications in a range of neurological disorders characterized by the involvement of MAPK signaling pathways. These conditions may include stroke, AD, and glaucoma.[100]

6.11 | Extracellular signal-regulated kinases

One of the potential mechanisms through which citicoline may exert its vascular effects is by influencing the ERK signaling pathway.[101] ERK is a protein kinase that holds a central role in governing various cellular functions, encompassing proliferation, differentiation, migration, and cell survival. ERK becomes activated through phosphorylation in response to a variety of triggers, including growth factors, cytokines, and stress-inducing factors. Once activated, ERK can stimulate angiogenesis by upregulating the expression of angiogenic factors and facilitating the migration and tube formation of endothelial cells. These specialized cells compose the inner lining of blood vessels.[102]

Numerous studies have provided evidence that citicoline has the ability to stimulate ERK phosphorylation in various cell types, including neurons, astrocytes, and endothelial cells. Notably, citicoline has been observed to induce angiogenesis in human brain microvessel endothelial cells (hCMEC/D3), a cell line designed to replicate the properties of the BBB. This suggests that citicoline's influence on the ERK pathway could play a role in promoting angiogenesis and may have implications for various physiological and pathological processes, particularly those involving the brain and its vasculature.[103] Citicoline was found to enhance various aspects of endothelial cell behavior in vitro, including wound healing, tube formation, and spheroid sprouting of hCMEC/D3 cells. Additionally, it increased the survival of these cells when exposed to staurosporin and calcium ionophore-induced apoptosis. These positive effects were attributed to citicoline's ability to induce ERK phosphorylation, as they were negated when an ERK inhibitor was used. This suggests that citicoline's promotion of endothelial cell activities and survival may be mediated through the activation of the ERK signaling pathway.[103]

Additionally, citicoline was observed to increase the expression of insulin receptor substrate-1 (IRS-1), a protein responsible for linking the insulin receptor to downstream signaling pathways, including the ERK pathway. Notably, when IRS-1 was suppressed using small interfering RNA (siRNA), it impeded the pro-angiogenic effects of citicoline on hCMEC/D3 cells. This underscores the crucial role of IRS-1 as a mediator in the process of citicoline-induced ERK activation and the consequent promotion of angiogenesis (Figure 4).[103,104] The exact mechanism by which citicoline regulates IRS-1 expression is not clear, but it may involve the modulation of phosphatidylcholine synthesis or membrane fluidity.[105,106]

6.12 | c-Jun N-terminal kinases

JNKs are enzymatic proteins that undergo activation in response to various stress-inducing stimuli, including cytokines, growth factors, oxidative stress, unfolded protein responses, and amyloid-beta peptides. Upon activation, JNKs instigate the phosphorylation and subsequent activation of c-Jun, a pivotal transcription factor accountable for regulating the expression of genes crucial in processes like inflammation, apoptosis, and neurodegeneration.[107] JNKs have been implicated in the etiology and advancement of a diverse spectrum of diseases, encompassing cancer, diabetes, obesity, cardiovascular ailments, and neurodegenerative disorders. The mismanagement of JNK signaling can significantly contribute to the onset and worsening of these maladies by perturbing critical cellular functions like inflammation, apoptosis, and stress responses.[108–110] Certainly, JNKs have been unequivocally shown to occupy a central role in the neural deterioration linked to AD. They actively contribute to the pathogenesis of AD by amplifying the generation of amyloid-beta, fostering hyperphosphorylation and aggregation of tau proteins, and instigating disruptions in mitochondrial function and oxidative stress. These manifold mechanisms firmly establish JNKs as a substantial contributor to the neurodegenerative processes observed in AD.[111]

It was previously reported that citicoline can effectively inhibit the activation of JNK and consequently prevent JNK-mediated neuronal death in focal cerebral ischemia/reperfusion injury model.[112] For instance, citicoline has been shown to reduce JNK phosphorylation and the expression of c-Jun in rat hippocampal neurons subjected to oxygen–glucose deprivation and reoxygenation (OGD/R). These findings suggest that citicoline's ability to modulate JNK signaling may contribute to its neuroprotective effects in conditions involving ischemic injury and neurodegeneration, such as AD.[112] Additionally, citicoline has been observed to reduce JNK activation and inhibit neuronal apoptosis in mouse models of transient global cerebral ischemia (tGCI). This further highlights citicoline's potential as a neuroprotective agent in the context of ischemic brain injury by targeting JNK-mediated cell death pathways.[113] Indeed, the cumulative evidence suggests that citicoline's neuroprotective effects may be attributed, at least in part, to its modulation of JNK signaling, thereby mitigating JNK-induced neuronal injury.[114]

6.13 | p38 MAPKs

Citicoline exerts anti-inflammatory effects via modulation of p38 mitogen-activated protein kinases (p38 MAPKs). P38 MAPKs are a family of stress-activated enzymes that control inflammation, apoptosis, and cell survival. p38 MAPKs have four isoforms (α, β, γ, and δ), each with distinct functions and substrates. p38 MAPKs are activated by numerous stimuli, such as cytokines, oxidative stress, and environmental toxins, and perform an important role in mediating neuroinflammation and neurodegeneration.[115]

Ischemic stroke ranks among the foremost global causes of both mortality and disability. This condition arises when a blood vessel responsible for delivering oxygen and essential nutrients to the brain becomes obstructed, ultimately leading to the demise of neurons and harm to brain tissue. An important outcome of this ischemic event is the stimulation of microglia and astrocytes, the brain's resident immune cells. These cells, in response, generate pro-inflammatory cytokines and ROS, intensifying the damage inflicted upon neurons.[116] Numerous studies have provided evidence that citicoline has the potential to mitigate the extent of ischemic brain injury through various mechanisms, including the reduction of infarct size, enhancement of neurological function, and promotion of neurogenesis.[117]

The impact of citicoline on p38 MAPK signaling and neuroinflammation was examined using a transient MCAO (tMCAO) rat model, a frequently employed model for studying focal ischemia. The researchers observed that citicoline treatment notably suppressed the phosphorylation of p38 MAPK in both neurons and glial cells 24 h following tMCAO induction.[46] Citicoline also lowered pro-inflammatory cytokines (TNF-α, IL-1β, and IL-6) expression and iNOS in the ischemic cortex and striatum.[65] Moreover, citicoline decreased microglial activation and astrocytic proliferation in the peri-infarct area. These results suggest that citicoline can inhibit p38 MAPK activation and reduce neuroinflammation in ischemic stroke.[77]

AD stands as the predominant type of dementia, marked by a gradual deterioration in cognitive abilities and memory loss. The hallmark pathological characteristics of AD involve the accumulation of amyloid-β (Aβ) plaques and the existence of neurofibrillary tangles (NFTs), composed of hyperphosphorylated tau protein. Aβ is produced from amyloid precursor protein (APP) through the cleavage actions of β- and γ-secretases, while the phosphorylation of tau is intricately controlled by an array of kinases and phosphatases. Both Aβ and tau exert deleterious effects on neuronal function and viability by disrupting crucial processes such as synaptic transmission, mitochondrial function, axonal transport, and autophagy.[118] Autophagy is a cellular mechanism responsible for breaking down impaired organelles and misfolded proteins through lysosomal degradation. It plays a crucial role in upholding neuronal equilibrium and thwarting protein aggregation. Nevertheless, in AD, autophagy becomes compromised due to lysosomal dysfunction, leading to the accumulation of Aβ and tau proteins.[119] Citicoline has been reported to improve cognitive function and reduce Aβ levels in patients with mild to moderate AD.[120]

A recent study by Liu et al. (2019) investigated the citicoline effect on tau phosphorylation and autophagy in a mouse model of AD induced by intracerebroventricular injection of Aβ25-35 peptide.[121] Moreover, citicoline treatment enhanced autophagy activity, as indicated by increased levels of light chain protein 3-II (LC3-II) and decreased levels of p62, two markers of autophagic flux. Citicoline also increased Beclin-1 and Atg5 expression, two key proteins involved in autophagy initiation.[122] It was suggested that citicoline may exert its beneficial effects on tau pathology and autophagy by modulating the mammalian target of rapamycin (mTOR) signaling pathway, which is a major regulator of protein synthesis and degradation.[123]

Citicoline has shown a potential therapeutic role in AD, especially in targeting tau pathology and autophagy dysfunction.[124] However, more studies are needed to investigate the findings in other animal models and human subjects and to reveal the molecular mechanisms underlying citicoline effects on tau phosphorylation and autophagy. Additionally, it would be interesting to explore whether citicoline has any synergistic effects with other drugs that modulate autophagy, such as rapamycin, lithium, or metformin, which have been shown to lessen tau levels and toxicity in several models of tauopathy.[125–127] Citicoline may represent a promising candidate for AD treatment, as it has a favorable safety profile and is widely available as a dietary supplement.

6.14 | Citicoline and STAT pathway

Citicoline holds promise in its ability to modulate the Janus kinases (JAK)/signal transducer and activator of transcription (STAT) signaling pathway, which has been implicated in a wide array of cellular processes, including but not limited to inflammation, apoptosis, and neurogenesis.[128] The activation of the JAK/STAT signaling pathway is triggered when cytokines and growth factors bind to specific cell surface receptors. This binding event initiates a series of sequential events, beginning with the phosphorylation and activation of JAK kinases. Subsequently, this activation leads to the phosphorylation and activation of STAT proteins. These activated STAT proteins then join together to form dimers and translocate into the cell nucleus, where they take charge of regulating the expression of specific target genes.[129] The JAK/STAT signaling pathway is meticulously regulated by a diverse array of control proteins, including suppressors of cytokine signaling, protein inhibitors of activated STATs, and protein tyrosine phosphatases. These regulatory proteins play an indispensable role in overseeing the initiation, duration, and ultimate termination of the signaling cascades.[130] Citicoline has been extensively documented to exert regulatory influence on the JAK/STAT signaling pathway through several mechanisms. Importantly, citicoline has the capacity to inhibit the activation of JAK2 and STAT3, leading to a decrease in the production of pro-inflammatory cytokines and the alleviation of oxidative stress.[131]

6.15 | Cholinergic anti-inflammatory pathway

One potential mechanism of action for citicoline involves the modulation of the cholinergic anti-inflammatory pathway (CAP). This neural circuit contributes to the regulation of inflammation by operating through the vagus nerve and the alpha7 nicotinic acetylcholine receptor (α7nAChR), which is found in immune cells. The CAP is initiated by Ach, the primary neurotransmitter of the parasympathetic nervous system. When Ach binds to the α7nAChR, it triggers the suppression of pro-inflammatory cytokine production.[132,133] This pathway has been implicated in the pathophysiology of various inflammatory diseases, making it a promising therapeutic target. Citicoline may enhance the CAP by increasing the availability of Ach and/or by directly activating α7nAChR. Citicoline is metabolized into choline and cytidine, and choline can be converted into Ach by ChAT in cholinergic neurons.[81] Therefore, citicoline may increase Ach levels by increasing the expression of ChAT and α7nAChR, as well as the release of Ach from nerve terminals. In addition, phosphatidylcholine is believed to be an α7nAChR agonist.[15] By stimulating the CAP, citicoline may reduce inflammation and oxidative stress, which are major contributors to neuronal damage and cognitive impairment.[65] Citicoline may also improve neurogenesis, synaptic plasticity, and neurotransmission, all of which are essential for learning and memory (Figure 4).

6.16 | Effect on apoptosis

Apoptosis is a highly regulated process that involves specific morphological features and biochemical cascades, such as caspase activation and DNA fragmentation. Apoptosis can be triggered by various factors, such as oxidative stress, inflammation, ischemia, and neurotoxicity. Citicoline can modulate apoptosis by affecting several pathways, such as phospholipid homeostasis,[134] mitochondrial dynamics,[135] cholinergic and dopaminergic transmission,[1] and proteasome activity.[81]

Numerous studies have provided evidence of citicoline's anti-apoptotic properties across various neurodegenerative models. For instance, in a model of chronic ocular hypertension, which simulates glaucoma, citicoline has been shown to decrease the number of apoptotic cells (as detected by TUNEL staining) and reduce the levels of caspase-3 and caspase-9 in the retinas of rats.[136] Furthermore, in a diabetic retinopathy model involving diabetic rats, citicoline was observed to reduce the expression of pro-apoptotic proteins, specifically Bax and Bad. Simultaneously, it increased the expression of anti-apoptotic proteins, such as Bcl-2 and Bcl-xL, within the retinal tissue.[137] Furthermore, in a model of stroke using rats subjected to transient focal ischemia, citicoline exhibited a protective effect by reducing the activation of caspase-8 and caspase-3, along with the cleavage of poly (ADP-ribose) polymerase, a marker indicative of DNA damage, within the brain tissue[138] (Figure 4). In these experimental models, citicoline not only effectively inhibited apoptosis but also exhibited the ability to enhance the expression of neurotrophic factors, including BDNF and ciliary neurotrophic factor. Furthermore, it promoted the regeneration of neurites, underscoring its potential as a neuroprotective and regenerative agent.[139]

6.17 | Effect on autophagy

One of the possible mechanisms of action of citicoline is to modulate autophagy, a cellular process that regulates the degradation and recycling of damaged or unwanted components. Autophagy is essential for maintaining cellular homeostasis and preventing oxidative stress, inflammation, and apoptosis. However, excessive or defective autophagy can also contribute to neuronal death and neurodegeneration.[140] Therefore, the balance between autophagy induction and inhibition is crucial for neuronal health.

Numerous studies have proposed that citicoline might augment autophagy in the brain by influencing various signaling pathways. One such mechanism involves the activation of AMP-activated protein kinase (AMPK), a pivotal regulator of both energy metabolism and autophagy. Citicoline appears to achieve this by stimulating Sirtuin1, a protein associated with cellular regulation and longevity.[61,141] Additionally, citicoline has been suggested to inhibit the mTOR, a protein complex that normally represses autophagy in the presence of ample nutrients. Furthermore, citicoline may elevate the levels of beclin-1, a protein that plays a pivotal role in initiating the formation of autophagosomes, the vesicles responsible for engulfing and breaking down cellular waste materials. These actions collectively support the notion that citicoline can facilitate and enhance autophagic processes in the brain.[142]

7 | CLINICAL RELEVANCE AND APPLICABILITY OF CITICOLINE AS AN ANTI-INFLAMMATORY AGENT

7.1 | Citicoline for acute ischemic stroke

Acute ischemic stroke stands as a prominent global cause of both mortality and disability. This condition arises when a blood vessel responsible for delivering vital oxygen and nutrients to the brain becomes obstructed, typically due to a clot or the presence of an atherosclerotic plaque. This blockage leads to tissue damage and the eventual demise of neurons in the affected area.[143] Inflammation is a key player in the mechanisms underlying ischemic stroke. It contributes significantly to the enlargement of the infarcted region, disruption of the BBB, and the activation of microglia and astrocytes. Furthermore, inflammation hinders the recovery process by inhibiting important processes such as neurogenesis, angiogenesis, and synaptic plasticity.[144]

Extensive research has been conducted on the potential use of citicoline as a treatment for acute ischemic stroke, with studies considering its efficacy both as a standalone therapy and in conjunction with thrombolysis or endovascular interventions. A meta-analysis, incorporating data from 13 randomized controlled trials (RCTs) involving 2398 patients who experienced acute ischemic stroke, demonstrated that citicoline was linked to a noteworthy reduction in mortality and disability after 3 months, in comparison to a placebo.[145] Furthermore, a separate meta-analysis that included data from 10 RCTs encompassing 2264 patients diagnosed with acute ischemic stroke found that citicoline was linked to a substantial enhancement in neurological outcomes after 12 weeks, in contrast to a placebo.[146] Additionally, a more specific subgroup analysis focusing on 4 RCTs involving 1372 patients who received intravenous thrombolysis for acute ischemic stroke demonstrated that citicoline was correlated with a noteworthy decrease in symptomatic intracranial hemorrhage and mortality when compared to a placebo.[147]

The positive impacts of citicoline in cases of acute ischemic stroke are likely attributable to its anti-inflammatory properties. Studies have suggested that citicoline has the ability to reduce the levels of pro-inflammatory cytokines, such as IL-6 and TNF-α, in both the serum and cerebrospinal fluid of individuals who have suffered from acute ischemic stroke.[148] Citicoline also reduces the expression of adhesion molecules, such as ICAM-1 and VCAM-1,[149] on leukocytes and endothelial cells, thus reducing inflammatory cells infiltration into the ischemic brain. Additionally, citicoline inhibits NF-κB activation.[150]

7.2 | Citicoline for TBI

TBI represents another prevalent global cause of both mortality and disability. It transpires when external forces inflict harm upon brain tissue, leading to the initiation of primary and secondary injury mechanisms. Primary injury encompasses the mechanical damage resulting from impact, penetration, or acceleration-deceleration forces. Conversely, secondary injury pertains to delayed processes following the primary injury, including inflammation, oxidative stress, excitotoxicity, apoptosis, and edema. In this context, inflammation assumes a central role in secondary injury, exacerbating neuronal damage and impeding the process of neurorepair.[151]

Citicoline possesses a range of biochemical, pharmacological, and pharmacokinetic characteristics that render it a promising candidate for mitigating the effects of TBI and promoting the restoration of brain function (as summarized in Table 2). Extensive research has explored citicoline's potential as a treatment for TBI, with investigations conducted in both animal models and human clinical trials. The findings suggest that citicoline holds promise in enhancing recovery by positively impacting consciousness, independence, and memory in TBI patients, while also demonstrating the ability to reduce brain edema and maintain the integrity of the BBB.[67] Numerous clinical trials have scrutinized the effectiveness and safety of citicoline when administered during the acute phase of TBI, with treatment initiation typically occurring within 24 h of the injury. A meta-analysis that amalgamated data from 11 such studies, encompassing a total of 2771 patients, revealed that citicoline was linked to a notably higher rate of independence among patients at the conclusion of the follow-up period, in comparison to those who received a placebo or standard care.[152] The studies did not reveal any significant impacts on mortality or adverse events. Importantly, the outcomes remained consistent regardless of the dose and route of citicoline administration. Consequently, citicoline emerges as a potentially beneficial intervention in the management of patients during the acute phase of TBI, offering both neuroprotection and neurorestoration. The anti-inflammatory properties of citicoline are believed to contribute to its neuroprotective effects in TBI, as it has been shown to reduce the levels of pro-inflammatory cytokines including IL-6, IL-8, IL-10, TNF-alpha, and interferon-gamma.

7.3 | Citicoline for glaucoma

Glaucoma represents a collection of eye conditions characterized by the gradual deterioration of the optic nerve, ultimately resulting in vision loss and potential blindness.

TABLE 2 Clinical relevance and applicability of citicoline.

Disease	Key findings	References
Acute ischemic stroke	• Citicoline is associated with a significant reduction in mortality and disability at 3 months	[145]
	• Citicoline is associated with a significant improvement in neurological outcomes at 12 weeks	[146]
	• Citicoline is associated with a significant reduction in symptomatic intracranial hemorrhage and mortality	[147]
	• Citicoline reduces pro-inflammatory cytokines and adhesion molecules, inhibiting inflammation	[148–150]
Traumatic brain injury	• Citicoline improves recovery of consciousness, independence, and memory in TBI patients	[67]
	• Citicoline significantly increases the rate of independence in TBI patients	[152]
	• Citicoline is well-tolerated and has a potential benefit for functional recovery in ICH	[153]
Glaucoma	• Citicoline improves the quality of life in chronic OAG patients	[87]
	• Citicoline protects the retinal nerve fiber layer and may slow down visual field changes	[134,154]
Multiple sclerosis	• Citicoline has beneficial effects on demyelination and remyelination in animal models of MS	[155–157]
	• Citicoline improves VEPs and retinal nerve fiber layer thickness in MS patients	[87,158]
Alzheimer's disease	• Citicoline supplementation improves cognitive performance and reduces inflammation in AD patients	[159,160]
Parkinson's disease	• Citicoline may improve motor function, delay cognitive impairment, and reduce levodopa dosage in PD	[66,161,162]
COVID-19-related complications	• Citicoline may have neuroprotective and cognitive-enhancing effects in COVID-19-related complications	[163]
Bipolar disorder	• Citicoline reduces inflammation and improves depressive symptoms and cognitive function in bipolar disorder	[164]
Major depressive disorder	• Citicoline reduces inflammation and improves depressive symptoms and cognitive function in MDD	[165,166]

Typically, this condition stems from increased IOP, which exerts pressure on the optic nerve head and impedes blood circulation. This, in turn, sets off a chain of events leading to neuroinflammation, oxidative stress, and the programmed cell death of retinal ganglion cells, the specialized neurons responsible for transmitting visual information from the eye to the brain.[167]

Citicoline can modulate several pathways involved in glaucoma pathogenesis, such as phospholipid metabolism, glutamate excitotoxicity, mitochondrial function, and Ach neurotransmission (Table 2). Citicoline can also enhance blood flow to the optic nerve and retina, and reduce IOP by increasing aqueous humor outflow.[168]

Numerous clinical investigations have assessed the impact of citicoline on glaucoma, employing both oral solutions and eye drops as modes of administration. In a recent international, multicenter, randomized, placebo-controlled crossover trial, it was noted that daily consumption of a 500 mg dose of citicoline oral solution resulted in an improvement in the quality of life for individuals suffering from chronic OAG. This enhancement in quality of life was assessed and quantified using the Visual Function Questionnaire-25 (VFQ-25).[87] Another study in rats showed that citicoline oral solution reduced vision loss and restored optic nerve signals without lowering IOP.[169] A pilot study in humans suggested that citicoline eye drops of 2% could slow down the progression of visual field changes and protect the retinal nerve fiber layer.[134,154] Nevertheless, it is important to note that these findings should be validated through more extensive and extended trials. Citicoline could potentially serve as a supplementary therapy for individuals with glaucoma, particularly those experiencing a diminished quality of life related to their vision or those who exhibit ongoing deterioration despite undergoing treatments to lower IOP. Citicoline's potential lies in its ability to introduce a new mechanism of action that directly addresses the neurodegeneration of the optic nerve.

7.4 | Citicoline for acute intracerebral hemorrhage

Intracerebral hemorrhage (ICH) is a form of stroke that results from bleeding within the brain tissue. It is linked to high rates of both mortality and morbidity, and currently, there is no targeted treatment available to enhance the outlook for affected patients. In the pathophysiology of intracerebral hemorrhage, inflammation plays a significant role, exacerbating issues such as edema, neuronal loss, and secondary injury.[170]

Citicoline has demonstrated neuroprotective and neuroregenerative capabilities in a variety of experimental models involving both ischemic stroke and ICH, as outlined in Table 2. Citicoline holds the potential to limit the extent of brain damage, foster neuroplasticity and neurogenesis, and contribute to improved functional recovery following a stroke.[171] Numerous clinical trials have explored the safety and effectiveness of citicoline in patients dealing with acute ischemic stroke, and they have generally reported positive results regarding neurological and functional outcomes. However, the available evidence for citicoline in the context of ICH is more limited and has yielded inconclusive findings.

For example, in a pilot study conducted by Secades et al. in 2006, 38 patients who had supratentorial ICH were randomly assigned to receive either a placebo or 1 g of citicoline every 12 h for a duration of 2 weeks. The primary objective of the study was to assess the number of adverse events, while the secondary aim was to determine the percentage of patients who achieved a modified Rankin Score (mRS) of 2 or less at the 3-month mark. The study revealed no significant disparity in the occurrence of serious adverse events between the two groups, indicating that citicoline was well tolerated among the participants. However, it is important to note that this study represents just one piece of the overall evidence, and further research is needed to establish citicoline's role in ICH management.[172] Furthermore, the study identified a favorable trend regarding functional independence, with five patients in the citicoline group achieving an mRS score of 2 or lower, as opposed to just one patient in the placebo group. However, it is worth noting that this difference did not reach statistical significance, and the study was not designed with the statistical power to detect such an effect. Consequently, the authors concluded that while citicoline appeared to be a safe treatment for human ICH with the potential for improving functional recovery, further validation of these findings was required through a larger-scale clinical trial.

Current clinical investigations into the impact of citicoline on acute ICH are limited and ongoing. One noteworthy study in progress is the CO-OPERA trial (Citicoline Oral Plus Endovascular Recanalization for Acute Ischemic Stroke). This trial is designed to assess the efficacy and safety of administering oral citicoline in conjunction with endovascular therapy, as compared to endovascular therapy alone, for patients experiencing acute ischemic stroke resulting from large vessel occlusion.

Additionally, the CO-OPERA trial intends to conduct a subgroup analysis on patients who develop hemorrhagic transformation following endovascular therapy. Hemorrhagic transformation is a complication that shares some similarities with ICH in terms of its pathophysiology and prognosis. The study aims to enroll 600 patients and is expected to conclude in 2023.[173]

7.5 | Citicoline for MS

MS is a chronic autoimmune disorder that primarily impacts the CNS, causing inflammation, demyelination, axonal loss, and neurodegeneration. MS is characterized by relapses and remissions of neurological symptoms, as well as progressive disability over time. The current treatments for MS are mainly immunomodulatory or immunosuppressive agents that aim to reduce the frequency and severity of relapses but have limited effects on neuroprotection and repair.[174]

Citicoline holds promise as a potential candidate for treating MS because it has demonstrated beneficial effects on both demyelination (the loss of the protective myelin sheath around nerve fibers) and remyelination (the restoration of myelin) in preclinical models of MS (Table 2).[175] Citicoline was evaluated in two complementary mouse models of MS: the experimental autoimmune encephalomyelitis (EAE) model and the cuprizone-induced myelin toxicity model. These models help researchers simulate and study different aspects of MS pathology and treatment.[155,156] The EAE model is induced by immunization with myelin antigens, which closely mimics the autoimmune aspect of MS. On the other hand, the cuprizone-induced myelin toxicity model involves the use of cuprizone, a toxin that selectively destroys oligodendrocytes—the cells responsible for producing myelin in the CNS. This model effectively replicates the demyelinating aspect of MS. By utilizing both models, researchers can gain insights into different facets of MS pathogenesis and treatment.

The authors found that citicoline was effective in both models when given orally at a dose of 500 mg/kg daily. In EAE mice, citicoline reduced the severity of neurological symptoms and prevented axonal loss, especially when given preventively from the day of immunization.[176] In cuprizone-treated mice, citicoline enhanced the recovery of myelin after withdrawal of the toxin.[157] Importantly, citicoline did not affect the inflammatory response or the immune cell infiltration in either model, suggesting that it acts by a different mechanism than conventional DMTs.

Citicoline's mechanism of action in MS is not totally understood, however, it may involve several aspects. First, citicoline may increase the availability of phospholipids for myelin synthesis and repair, as well as for neuronal membrane integrity and function.[177] Second, citicoline may modulate various enzyme expression and activity and factors involved in myelin formation and maintenance, such as ChAT, acetylcholinesterase, nerve growth factor, cAMP response element-binding protein, and BDNF.[58,177–179] Third, citicoline may provide protection to neurons from oxidative stress and excitotoxicity by elevating the levels of GSH, an important antioxidant, while simultaneously reducing the levels of glutamate, a neurotransmitter associated with excitotoxicity.[180] Fourth, citicoline may improve mitochondrial function and energy metabolism in both neurons and oligodendrocytes.[177]

The evidence from animal studies suggests that citicoline may have a potential role as an adjunct therapy for MS, especially for enhancing remyelination and neuroprotection. However, clinical trials are needed to confirm its efficacy and safety in humans. There are some indications that citicoline may be beneficial for MS patients based on its effects on visual function. Citicoline has been shown to improve visual evoked potentials (VEPs), which are electrical signals generated by the brain in response to visual stimuli. VEPs are often used as a marker of optic nerve function and demyelination in MS patients. Citicoline may improve VEPs by enhancing retinal function and optic nerve conduction.[158]

In a recent pilot study conducted by Rossetti et al., the impact of oral citicoline was investigated. This study involved 41 patients with relapsing–remitting MS who had previously experienced optic neuritis. Over a period of 12 weeks, these patients received a daily dose of 500 mg of citicoline. The study's findings indicated that citicoline brought about significant improvements in VEP, with notable enhancements observed in both latency and amplitude. Additionally, optical coherence tomography measurements revealed favorable changes in the thickness of the retinal nerve fiber layer and ganglion cell layer, when compared to a placebo group.[87] These results suggest that citicoline may have a positive effect on optic nerve function and structure in MS patients, possibly reflecting its remyelinating and neuroprotective properties.

7.6 | Citicoline for Alzheimer's disease

Previous studies have shown that citicoline may help reduce inflammation and improve cognitive function in AD patients by modulating the activity of microglia (Table 2). In a rigorously conducted randomized, double-blind, placebo-controlled trial that encompassed 60 patients diagnosed with mild to moderate AD, the administration of citicoline as a supplement (at a dose of 500 mg twice daily for a duration of 12 weeks) yielded notable benefits. Specifically, citicoline significantly improved cognitive performance and concurrently reduced markers of inflammation in both plasma and cerebrospinal fluid when compared to a placebo. This study underscores the potential of citicoline as a therapeutic intervention for individuals with AD.[160] Furthermore, in another well-designed randomized, double-blind, placebo-controlled trial that included 40 patients diagnosed with mild to moderate AD, the administration of citicoline as a supplement (at a dosage of 500 mg taken twice daily over a period of 9 months) demonstrated substantial advantages. Specifically, citicoline supplementation significantly enhanced memory and attention while concurrently reducing oxidative stress and inflammation markers in plasma when compared to a placebo. These findings provide additional support for the potential therapeutic role of citicoline in managing AD.[159]

7.7 | Citicoline for Parkinson's disease

Citicoline has demonstrated its neuroprotective qualities and cognitive enhancement potential in various neurological conditions, including PD, as summarized in Table 2. PD is a progressive neurodegenerative disorder characterized by motor symptoms like tremors, rigidity, bradykinesia, and postural instability, as well as non-motor symptoms such as cognitive impairment, mood disorders, and sleep disturbances. The primary pathological hallmark of PD is the degeneration of dopaminergic neurons in the substantia nigra, leading to reduced dopamine levels in the striatum and other brain regions. The standard treatment for PD is levodopa, a dopamine precursor capable of crossing the BBB and being converted into dopamine by the remaining dopaminergic neurons.[181] Indeed, levodopa, while effective in treating PD symptoms, comes with its share of limitations, including the development of wearing-off effects, dyskinesias (abnormal involuntary movements), and potential adverse effects on other neurotransmitter systems. This underscores the pressing need for alternative or supplementary therapeutic approaches that can enhance symptom management and overall quality of life for individuals living with PD.[182]

Citicoline holds promise as a potential adjuvant therapy for PD through various mechanisms. It may increase dopamine synthesis and inhibit dopamine reuptake in the brain, potentially contributing to improved dopamine levels. Additionally, citicoline has the potential to enhance the efficacy of levodopa, allowing for the reduction of its dosage and thereby potentially mitigating its side effects. Beyond its impact on motor symptoms, citicoline may also have a positive influence on the cognitive function and mood of individuals with PD. These multifaceted effects make citicoline an intriguing candidate for PD management.[14]

A notable study employed a randomized approach, enrolling 85 patients diagnosed with primary PD. These patients were allocated into two groups: one received their regular dose of levodopa in combination with 1200 mg of citicoline daily, while the other group received half of their typical levodopa dose alongside the citicoline supplement. The outcomes of this study revealed that citicoline as an adjuvant therapy yielded several favorable effects, including improvements in motor function, a delay in the progression of cognitive impairment, and a reduction in symptoms such as bradykinesia, rigidity, and tremors in individuals with PD. Additionally, citicoline exhibited a levodopa-sparing effect, enabling the postponement of the initiation of levodopa treatment and a reduction in the required dosage of levodopa. These findings underscore the potential benefits of citicoline as a supplementary therapy for PD management.[161]

An additional study has demonstrated the efficacy of citicoline in the treatment of PD when administered intravenously. This treatment approach has also proven effective in addressing levodopa-induced pharmacotoxic psychoses, which can include symptoms like confusion and paranoid hallucinatory states, often occurring during the later stages of the disease. Importantly, these therapeutic benefits were achieved with minimal adverse effects, highlighting citicoline's potential utility in managing PD-related symptoms and complications.[183] Additionally, a systematic review that assessed the effectiveness of citicoline as adjunctive therapy for improving symptoms in PD reported promising findings. However, the review also emphasized the necessity for larger-scale clinical trials to establish the optimal dosage and treatment duration, further advancing our understanding of citicoline's potential role in PD management.[66]

7.8 | Citicoline for COVID-19-related cognitive and other neurologic complications

COVID-19 is a viral infection that can cause severe respiratory distress and multi-organ failure. COVID-19 can also affect the nervous system and cause neurological complications, such as headache, dizziness, loss of smell and taste, confusion, delirium, seizures, stroke, and encephalitis. Some of these complications may persist after recovery from the acute infection and result in long-term cognitive impairment and neurological disability.[52] Citicoline may be a potential adjunctive therapy for preventing and treating COVID-19-related cognitive and other neurologic complications (Table 2). Citicoline has anti-viral, anti-inflammatory, neuroprotective, and neurorestorative properties that may help counteract the detrimental effects of COVID-19 on the brain. Citicoline may also enhance the synthesis of Ach, a neurotransmitter that is involved in memory, attention, and learning.[65] A recent review article suggested that citicoline may be beneficial for COVID-19 patients with neurological symptoms by reducing viral replication, modulating immune responses, protecting brain cells from damage and death, promoting neural repair and regeneration, and improving cognitive function.[163]

7.9 | Citicoline for bipolar disorder

Bipolar disorder is a mental health condition characterized by recurrent episodes of mania (marked by elevated mood, increased energy, and heightened activity levels) and depression (characterized by low mood, reduced energy, and decreased activity levels). This disorder is linked to chronic inflammation within the brain, which could potentially contribute to mood fluctuations, cognitive deficits, and an elevated risk of neurodegenerative conditions.[184] Citicoline appears to hold promise as an adjunctive treatment for bipolar disorder by virtue of its potential to alleviate brain inflammation and enhance mood and cognitive functioning, as indicated in Table 2. In support of this, a randomized controlled trial involving 40 individuals diagnosed with bipolar disorder demonstrated that supplementation with citicoline (at a dosage of 500 mg taken twice daily over a span of 12 weeks) led to a significant reduction in plasma levels of IL-6, a pro-inflammatory cytokine that is typically elevated in bipolar disorder. Additionally, citicoline brought about significant improvements in depressive symptoms and cognitive performance when compared to a placebo group. These findings suggest that citicoline may have a role in the management of bipolar disorder.[164]

7.10 | Citicoline for MDD

Major depressive disorder (MDD) is a common mental health condition characterized by persistent feelings of sadness, a diminished interest in once-enjoyed activities, reduced self-esteem, disrupted sleep patterns, fatigue, and, in severe instances, contemplation of self-harm or suicide. It is worth noting that MDD is associated with elevated inflammation in the brain, which has the potential to disrupt neurotransmission, synaptic plasticity, neurogenesis, and the expression of neurotrophic factors. In light of this, citicoline may emerge as a valuable adjunctive therapy for MDD due to its capacity to mitigate brain inflammation and augment neurotransmitter synthesis and function, offering a potential avenue for improving the condition[165] (Table 2). A randomized controlled trial that enrolled 121 individuals diagnosed with MDD yielded noteworthy results. In this trial, participants received citicoline supplementation at a dosage of 500 mg taken twice daily over a duration of 12 weeks. The study demonstrated that citicoline led to a significant reduction in plasma levels of CRP, which serves as a marker for systemic inflammation and is typically elevated in individuals with MDD. Additionally, citicoline had a positive impact by significantly improving both depressive symptoms and cognitive function in comparison to a placebo group. These findings suggest that citicoline could offer potential benefits as an adjunctive treatment for MDD.[185]

8 | CONCLUSION

In summary, citicoline has shown promise as a therapeutic agent for various neurological and inflammatory diseases. Its mechanisms of action are complex and context-dependent, including modulation of phosphatidylcholine synthesis, proteasome activity, JAK/STAT signaling pathway, ERK signaling, and transcription factors. Citicoline may exert its effects by balancing pro- and anti-inflammatory responses, enhancing neuroplasticity, inducing angiogenesis, and regulating proteostasis and energy metabolism. Supplementary studies are needed to elucidate the molecular details of citicoline's interactions with various pathways and its optimal dosage and timing for different clinical scenarios, especially STAT pathways. Revealing the mechanisms of action of citicoline will provide insights for the development of more effective therapies for neurological and inflammatory diseases.

Future studies should aim to elucidate the molecular pathways and the pharmacokinetics of citicoline in different brain regions and cell types, as well as its interactions with other neurotransmitters and neuromodulators. Furthermore, it is essential to conduct RCTs with larger sample sizes and more extended follow-up durations to establish both the effectiveness and safety of citicoline across diverse patient populations and clinical scenarios. This would provide a more comprehensive understanding of the potential benefits and risks associated with citicoline as a therapeutic intervention. Additionally, biomarkers of response and predictors of outcome should be identified to optimize the personalized treatment of patients with citicoline. Finally, the potential synergistic effects of citicoline with other pharmacological or non-pharmacological interventions should be explored to enhance its therapeutic benefits.

AUTHOR CONTRIBUTIONS

Simona Cavalu, Mahmoud E. Youssef, Asmaa Ramadan, Sameh Saber, Elsayed A. Elmorsy, Rabab S. Hamad, and Mustafa Ahmed Abdel-Reheim contributed to the review article in conception, design, analysis, and interpretation of the data; the drafting of the paper, revising it critically for intellectual content, and the final approval of the version to be published. All authors agree to be accountable for all aspects of the work.

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DISCLOSURES

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data was created or analyzed in this study.

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